

Final Review Lecture STAT 286

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Dezember 8th, 2025

Outline

- 1 Observational Studies
- 2 Partial Identification
- 3 Difference-in-Differences
- 4 Synthetic Control

Tablets and Test Scores

- **Goal:** estimate causal effect of providing tablets on test scores

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- **Observational setting:** schools decide themselves whether to adopt tablets.
- **Confounding:** Wealthier schools are both
 - more likely to adopt tablets, and
 - have higher test scores (tutoring, staffing, etc.).
 - Confounding in treatment **and** outcome (one would be ok).
- We suspect that

$$\mathbb{E}[Y_i \mid T_i = 1] - \mathbb{E}[Y_i \mid T_i = 0] \neq \mathbb{E}[Y_i(1) - Y_i(0)].$$

Assumption I: Super-Population

- For each unit $i = 1, \dots, n$, we observe
 - covariates X_i ,
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- In the tablet example:
 - X_i : wealth characteristics of school i ,
 - T_i : indicator that school i adopts tablets,
 - $Y_i(t)$: test score that would be observed for school i if $T_i = t$.

Assumption II: SUTVA

Assumption 1 (SUTVA).

- *No interference:*

$$Y_i(T_1, \dots, T_n) = Y_i(T_i).$$

Tablet adoption of other schools don't affect school i . Student's with tablets don't move to schools without.

Assumption 1 (SUTVA).

- *No interference:*

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Tablet adoption of other schools don't affect school i . Student's with tablets don't move to schools without.

- *Consistency:*

$$Y_i = Y_i(T_i).$$

All tablets are treated the same. No adopted tablet is significantly different (more features etc.)

Assumption 2 (Strong Ignorability / Unconfoundedness).

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- Intuition: after conditioning on $\mathbf{X}_i$, there are **no unobserved confounders** that jointly affect T_i and Y_i .

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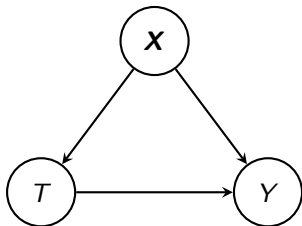
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Tablet example.

- Suppose $X_i = \text{Wealth}$
- Within each wealth level, schools with and without tablets differ only by treatment, not by other unobserved factors.

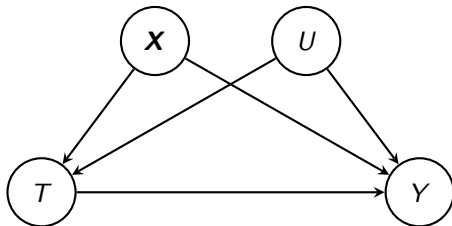
Ignorability DAG

Ignorability holds



No unmeasured confounding given X .

Ignorability violated

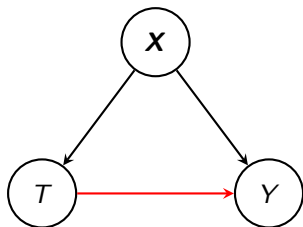


Unmeasured confounding given X .

- If U exists and affects both T and Y , then $\{Y(1), Y(0)\} \not\perp\!\!\!\perp T \mid X$; estimates that omit U are biased.
- Tablet example: U parental involvement

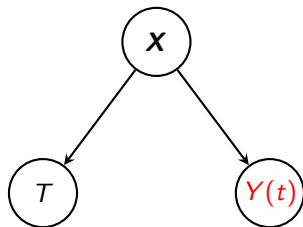
Observational / Causal DAG

Observational



T affects Y .

Causal



T affects $Y(t)$ only through X .

- By setting Y to level $T = t$ ($Y \rightarrow Y(t)$), we remove the arrow from T to Y .
- All effects of T to Y go through X . Conditioning *blocks* that path.
- Only condition on confounder, never on collider
- Rule of thumb: Arrow must go into Y .

Assumption 3 (Strong Overlap).

$$\exists \eta \in \left(0, \frac{1}{2}\right) \text{ s.t. } \eta < e(\mathbf{x}) = \Pr(T = 1 \mid \mathbf{X} = \mathbf{x}) \leq 1 - \eta \quad \text{for all } \mathbf{x}.$$

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Tablet example.

- With $X = \text{Wealth}$, strong overlap requires that at every wealth level there are both tablet and non-tablet schools.

Ignorability vs. Overlap: A Tension

In observational studies, we rely on both:

- **Ignorability:** $\{Y(1), Y(0)\} \perp\!\!\!\perp T \mid \mathbf{X}$.
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- **Overlap:** $\eta < \Pr(T = 1 \mid \mathbf{X}) \leq 1 - \eta$.
- Adding more covariates to $\mathbf{X}$
 - makes ignorability more plausible (less risk of omitted confounders),
 - but makes overlap less plausible: treatment becomes more predictable and $e(\mathbf{X})$ gets closer to 0 or 1.

Identification: ATE

Under these assumptions we have

$$\mathbb{E}[Y_i(1) - Y_i(0)]$$

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where $\mu_t(\mathbf{x}_i) = \mathbb{E}[Y_i \mid T_i = t, \mathbf{X}_i = \mathbf{x}_i]$. We then estimate

$$\hat{\tau}_{\text{ATE}}^{\text{reg}} = \frac{1}{n} \sum_{i=1}^n \hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i),$$

with a *consistent* $\hat{\mu}$.

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This identification strategy suggests estimating $\hat{\mu}_t$ and then forming the regression estimators

$$\hat{\tau}_{\text{ATT}}^{\text{reg}} = \frac{1}{n_1} \sum_{i=1}^n T_i (Y_i - \hat{\mu}_0(\mathbf{X}_i)).$$

Note that under an experimental design $\text{ATE} = \text{ATT}$.

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Partial Identification: Big Picture

- Identification relies on strong Assumptions.
- Often, researchers doubt unconfoundedness
- Can we still say something about ATE if unconfoundedness fails?

Setup: Binary ATE Without Randomization

- Binary treatment $T_i \in \{0, 1\}$ and binary outcome $Y_i \in \{0, 1\}$.
- Potential outcomes $Y_i(0)$, $Y_i(1)$ and the ATE

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- Assume:
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 - SUTVA,
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- Assume:
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 - Overlap: $0 < \Pr(T_i = 1) < 1$,
 - **No** randomization / unconfoundedness.
- Question: what can we still say about τ from observational (Y_i, T_i) alone?

Binary ATE: Decomposing the Problem

$$\begin{aligned}\mathbb{E}[Y_i(1)] &= \Pr(Y_i(1) = 1 \mid T_i = 1) \Pr(T_i = 1) + \Pr(Y_i(1) = 1 \mid T_i = 0) \Pr(T_i = 0), \\ \mathbb{E}[Y_i(0)] &= \Pr(Y_i(0) = 1 \mid T_i = 1) \Pr(T_i = 1) + \Pr(Y_i(0) = 1 \mid T_i = 0) \Pr(T_i = 0).\end{aligned}$$

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From (Y_i, T_i) using SUTVA and Overlap we can estimate

- $\Pr(T_i = t)$
- $\Pr(Y_i(1) = 1 \mid T_i = 1) = \Pr(Y_i = 1 \mid T_i = 1)$
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However, we don't know the “cross-world” terms

$$\Pr(Y_i(1) = 1 \mid T_i = 0), \quad \Pr(Y_i(0) = 1 \mid T_i = 1)$$

are not identified without randomization.

Binary ATE: Worst-Case Bounds

- **Idea:** Bound the probabilities

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- Plugging these into the decompositions yields bounds for each mean:

$$\Pr(Y_i = 1 \mid T_i = 1) \Pr(T_i = 1) \leq \mathbb{E}[Y_i(1)] \leq \Pr(Y_i = 1 \mid T_i = 1) \Pr(T_i = 1) + \Pr(T_i = 0),$$

$$\Pr(Y_i = 1 \mid T_i = 0) \Pr(T_i = 0) \leq \mathbb{E}[Y_i(0)] \leq \Pr(T_i = 1) + \Pr(Y_i = 1 \mid T_i = 0) \Pr(T_i = 0).$$

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- Combining worst-case upper and lower bounds yields

$$\tau_{\min} \leq \tau \leq \tau_{\max},$$

where

$$\tau_{\max} = \Pr(Y_i = 1 \mid T_i = 1) \Pr(T_i = 1) + \Pr(Y_i = 0 \mid T_i = 0) \Pr(T_i = 0),$$

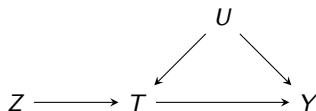
$$\tau_{\min} = -\Pr(Y_i = 0 \mid T_i = 1) \Pr(T_i = 1) - \Pr(Y_i = 1 \mid T_i = 0) \Pr(T_i = 0).$$

IV Setting: Setup and DAG

- Binary encouragement $Z_i \in \{0, 1\}$, binary treatment $T_i \in \{0, 1\}$, binary outcome $Y_i \in \{0, 1\}$.

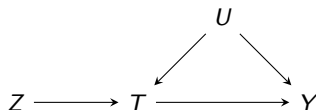
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- Goal: bounds on the ATE (not LATE!)

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$$

when T is confounded by U but Z is a valid instrument.

IV Setting: Latent Types and the Estimand

- Define the *latent type* for unit i :

$$U_i = (T_i(0), T_i(1), Y_i(0), Y_i(1)) \in \{0, 1\}^4,$$

so there are 16 possible types u .

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- The ATE can be written via the law of total probability:

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- So τ is a *linear function* of the unknown probabilities π_u .

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IV Setting: Sharp Bounds via a Linear Program

This leads to a pair of linear programs in the unknowns $\Pr(U_i = u)$:

$$\text{maximize / minimize } \tau = \sum_u (\mathbb{1}\{y_1(u) = 1\} - \mathbb{1}\{y_0(u) = 1\}) \Pr(U_i = u)$$

subject to

$$\Pr(Y_i = y, T_i = t, Z_i = z) = \sum_u \mathbb{1}\{y_t(u) = y, t_z(u) = t\} \Pr(Z_i = z) \Pr(U_i = u)$$
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How to solve this:

- 1 Derive bounds on τ within the constraints
- 2 Tip: make sure that the probability bounds are satisfied.
- 3 Show that these are *sharp*: there exists a probability combination of $\{\Pr(U_i = u)\}_u$ that achieves these bounds.

Outline

- 1 Observational Studies
- 2 Partial Identification
- 3 Difference-in-Differences**
- 4 Synthetic Control

Panel Data: Missing Outcome Problem

- We now observe the *same* units repeatedly over time $\{Y_{it}\}_{i=1,\dots,N,t=1,\dots,T}$.
- Interested in ATT
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- Interested in ATT
- All units start untreated; at some later time, a subset becomes treated
- For a newly treated unit i at time $t + 1$, the causal effect is

$$\tau_{it+1} = Y_{it+1}(1) - Y_{it+1}(0) = Y_{it+1} - Y_{it+1}(0)$$

- Need to impute: $Y_{it+1}(0)$

Naive Time Comparison: Before–After

- **Before–after idea:** compare the same unit over time:

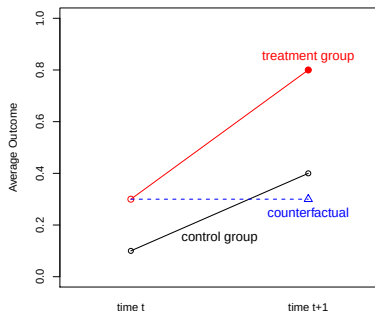
$$\hat{\tau}_{it+1} = Y_{it+1} - Y_{it}.$$

- This assumes that

$$Y_{it+1}(0) = Y_{it}(0),$$

i.e. the untreated outcome would not change over time.

- Unrealistic: ignores common trends, macro shocks, aging, etc.



Before–after comparison (taken from Kosuke Imai's lecture slides).

Naive Cross-Sectional Comparison

- **Cross-sectional idea** use units that remain in the control group at time $t + 1$.
- Compare treated unit i to a control unit j at the same time:

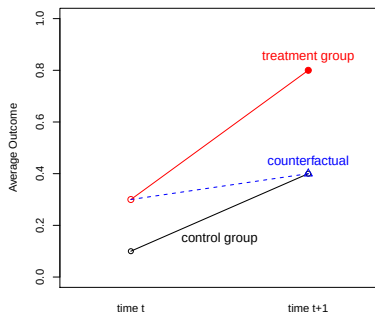
$$\hat{\tau}_{it+1} = Y_{it+1} - Y_{jt+1}.$$

- For treated unit i and control unit j , this assumes

$$Y_{it+1}(0) = Y_{jt+1}(0),$$

i.e. they would have had the same untreated outcome at $t + 1$.

- Unrealistic: treatment is not randomized, systematic differences between treated and control units (selection into treatment)



Cross-sectional comparison (taken from Kosuke Imai's lecture slides).

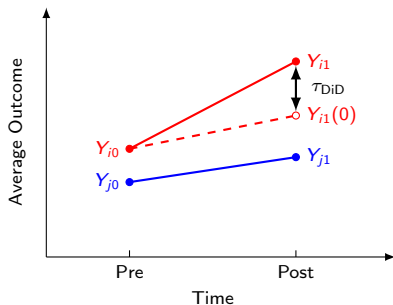
Difference-in-Differences: Combining Time and Units

- **Idea:** combine the before–after and cross-sectional comparisons.
- **Parallel trends assumption:**

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0)] = \mathbb{E}[Y_{j1}(0) - Y_{j0}(0)],$$

i.e. in the absence of treatment, treated and control units would follow the same average trend.

- $\tau_{\text{DiD}} = (Y_{i1} - Y_{i0}) - (Y_{j1} - Y_{j0})$.
- **Imputes:**
 $Y_{it+1}(0) = Y_{i0}(0) + (Y_{j1}(0) - Y_{j0}(0))$,
i.e. before treatment + trend of control



Difference-in-differences in the 2×2 case.

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- Causal estimand: the individual effect on unit N at time T ,

$$\tau_N = Y_{NT}(1) - Y_{NT}(0) = Y_{NT} - Y_{NT}(0).$$

- Problem: $Y_{NT}(0)$ is unobserved $\Rightarrow$ need to impute it.

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- We use the donor pool of control units $i = 1, \dots, N - 1$ to approximate $Y_{Nt}(0)$.

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- The treated unit's untreated path is a **time-invariant** weighted average of control paths.

Interpolation vs. Extrapolation

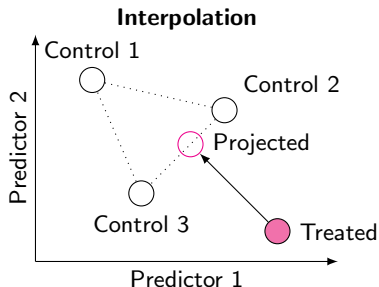
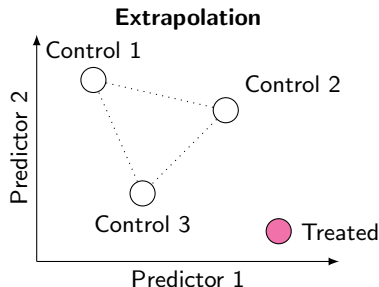
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- **Extrapolation:**
 - The treated unit lies *outside* the convex hull of controls.
 - Unconstrained regression can predict far outside the range of the data.
- **Interpolation:**
 - Project the treated unit into the convex hull of controls and predict *within* a plausible region.
 - This guards against wild predictions when we only care about a single treated unit.



Extrapolation (left) predicts using controls outside their convex hull; interpolation (right) projects into the

Interpolation vs. Extrapolation

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 - All controls have unemployment between 5% and 10%.
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 - All controls have unemployment between 5% and 10%.
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 - Synthetic control restricts predictions to [5%, 10%].
- As a prediction method for a single trajectory, avoiding extrapolation is crucial.

Fitting the Synthetic Control

- Choose $\hat{w} = (\hat{w}_1, \dots, \hat{w}_{N-1})$ to match the pre-treatment trajectory of unit N :

$$\hat{w} = \arg \min_w \sum_{t=1}^{T-1} \left(Y_{Nt} - \sum_{i=1}^{N-1} w_i Y_{it} \right)^2$$

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- Works best with long, stable pre-treatment trajectories and a large donor pool.

Important topics not discussed:

- RDD
- Double Robust Estimation
- Mediation

Any Questions?

Good luck!

You've got this! Go show off those causal superpowers.